

Solutions to weekly problems and exercises: Week 7b

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1 Week 7b: Exponents and Roots

1.1 Learning Goals

- Know how to evaluate exponential expressions
- Know how to write exponential expressions when multiplying or dividing with the same base.
- Know how to find roots.

1.2 Exponents

Rules:

1. $a^m \times a^n = a^{m+n}$

2. $\frac{a^m}{a^n} = a^{m-n}$

3. $(a^m)^n = a^{mn}$

4. $a^{-m} = \frac{1}{a^m}$

1.2.1 Problems

Simply:

1. $x^2 \times x^3 = \dots$

2. $\frac{3^5}{3^2} = \dots$

3. $(4^5)^2 = \dots$

4. $2^{-3} = \dots$

1.2.2 Watch

1.2.3 Exercises

Simply:

1. $z^3 \times z^2 = \dots$

2. $\frac{4^3}{4^2} = \dots$

3. $(5^2)^3 = \dots$

4. $3^{-2} = \dots$

1.3 Attempt the Exercises on Moodle

At this point, you would have worked on the above exercise questions.

Now, I want you to attempt these questions on Moodle.

Please click the link below:

[Week 7a Exercises on Moodle](#)

Once you submit all your answers on Moodle, you complete your learning on this week's topic.

1.4 Roots

Square roots. 4 is called a square root of 16 because $4^2 = 16$. The symbol $\sqrt[2]{}$ or $\sqrt{}$ is used to represent square root. For example, $\sqrt{16} = 4$. $\sqrt{16}$ can be written as $16^{\frac{1}{2}}$.

Cube roots. 2 is called a cube root of 8 because $2^3 = 8$. The symbol $\sqrt[3]{}$ is used to represent cube root. For example, $\sqrt[3]{8} = 2$. $\sqrt[3]{8}$ can be written as $8^{\frac{1}{3}}$.

** In general, $\sqrt[n]{a}$ can be written as $a^{\frac{1}{n}}$. It is hard sometimes to evaluate this expression manually. You can use a scientific calculator or **Excel** to do it. For example, by typing `=16^(1/4)` in **Excel**, it gives 2. That is $\sqrt[4]{16} = 16^{\frac{1}{4}} = 2$. You can prove that $2 \times 2 \times 2 \times 2 = 16$

1.4.1 Problems

Evaluate $\sqrt[3]{16}$ (you can use a scientific calculator or **Excel**)

1.4.2 Watch

1.4.3 Exercises

Evaluate

1. $\sqrt[3]{81}$
2. $\sqrt[4]{256}$
3. $\sqrt{81}$
4. $\sqrt[3]{216}$
5. $\sqrt[4]{625}$
6. $\sqrt[3]{15.63}$

1.5 Attempt the Exercises on Moodle

At this point, you would have worked on the above exercise questions.

Now, I want you to attempt these questions on Moodle.

Please click the link below:

[Week 7b Exercises on Moodle](#)

Once you submit all your answers on Moodle, you complete your learning on this week's topic.